TASKS TO BE DONE UNDER EACH EXPERIMENT

PRACTICAL EXAM - 25 Marks

15 Marks - Practical (10 [Execution] + 5 [Theory])
10 Marks - Oral

Sr. No	Experiments for Practical	Preparation Resource
1	Setup VM on Virtualbox	Lab Manual
2	Setup VM on KVM	Shared a command Pdf
3	Setup of Owncloud /Nextcloud	Lab Manual
4	IAAS - AWS EC2 [Any 2 tasks]	AWS Educate
5	STaaS - AWS S3 [Any 2 tasks]	AWS Educate
6	Make containers using Dockers	Shared a command Pdf
7	Paas - Host basic app on Render OR Vercel	**Render** **https://youtu.be/G13EOZ_pI1Y?si=uzoKI4aqrC-7JNtf** **Vercel** https://youtu.be/sSVl2JReVSg?si=XgFO9EylgG7QYyxu

1) Installation and configuration of KVM - A Type 1 Hypervisor to create and manage virtual machine and explore paravirtualization settings

- Make a VM as per given specification
- Do paravirtualization settings and show that it is enabled
- Make manual hard disk partitions while OS installation

2) Implement the concept of Full Virtualization by Configuring a virtual machine using VirtualBox - A Type 2 Hypervisor

- Make a VM as per given specification
- Ping base machine from VM

3) Implementation of a cloud storage server on local machine by using Own Cloud/ Nextcloud

Setup the Owncloud/Nextcloud Storage server

- Access it from any other device in the lab [over LAN or WAN(ngrok)].
- Show how you upload data onto the server and where it it stored in the filesystem

Note : Command list can be referenced during the exam.

4) Study and implementation of Infrastructure as a Service (IAAS) cloud service model using AWS EC2 (not to be performed practically)

- Write down the steps for doing any 3 tasks

5) Study and implementation of Storage as a Service (STaaS) cloud service model using AWS S3 (not to be performed practically)

- Write down the steps for doing any 3 tasks

6) Creating Containerized applications with Dockers and exploring DockerHub

- Make a dockerfile
- Create an image file
- Run the image file to make a container
- Push the image file onto the dockerHub
- Pull an image from dockerHub and run it on your computer

7) Deployment of a Web Application using Platform as a Service (PaaS) Environment

- Host a simple app on any online hosting platform like vercel/render/heroku,etc.